

# TRAVEL STYLE SEGMENTATION & DRIFT MONITORING

## ITCS355 - Machine Learning Operations

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# ONE PROFILE, A SIMPLE TRAVEL SEGMENT

A travel platform can use rating profiles to group similar interests.

### INPUT

24 category ratings [1]

Each value: 0-5

### MODEL

K-Means, k=5

Shared preprocessing

### OUTPUT

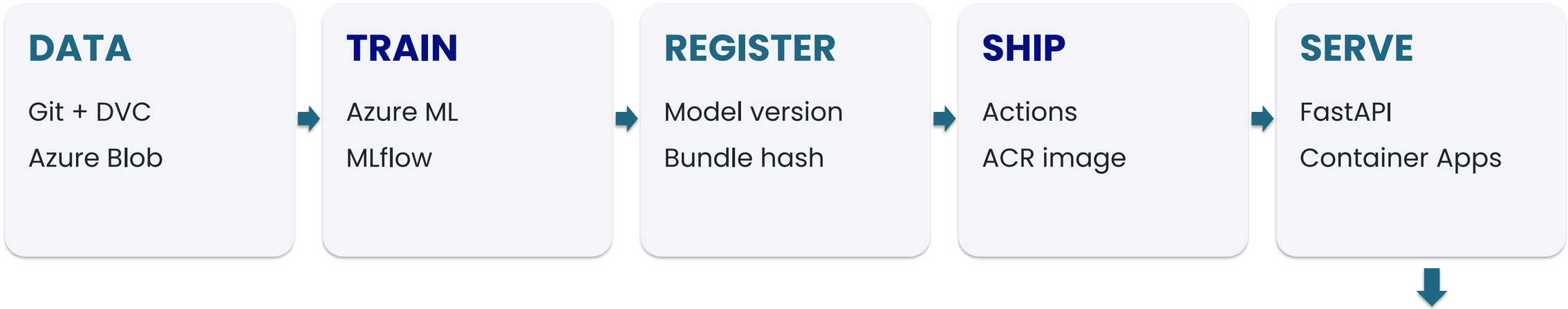
Segment + distance

Uncertainty + version

Segment names are provisional. This is a preference tool, not a personality test.

Source: docs/MODEL_CARD.md; reports/bundle-manifest.json; UCI [1]

# FROM VERSIONED DATA TO A MONITORED API




Tests gate deployment.

Failure checks feed back into tests.

Azure region: centralindia | Provider code: cloudlayer/

Source: README.md; docs/reference/course/project/project-brief.md [2]

# ONE VERSIONED DATASET, ONE REPEATABLE PIPELINE

**5,456**
Raw rows

**5,454**
Accepted rows

**2**
Quarantined rows




**CC BY 4.0 [1]** | Raw SHA-256 + DVC pointer | Seed 42

Train-only imputation/scaling. Identical profiles stay in one split.

Source: reports/data-audit.json; prepared-data.json; selection.json; UCI [1]

# THE SAME BUNDLE REACHES THE API

## MANAGED TRAINING

Azure ML job: Completed

Fresh-fit equivalence: passed

Registry: travel-style, version 1

## SERVING BUNDLE

Preprocessing + feature order

Centers + mapping + thresholds

Hash and golden checks at startup

## Model version: abfd88e893ca

Immutable images use one canonical bundle.

```
Bundle SHA-256: 71f3ba41b4b964f5...
CI image digest: f124df59007a...
```

Source: reports/finish-training-registry.json; bundle-manifest.json; ci-deploy-result.json

# LIVE DEMO: VALID AND UNEXPECTED INPUT

| Request | Expected / verified |
|---|---|
| Valid 24-field profile | 200 + segment |
| Missing fields / invalid rating | 422, clear validation error |
| Batch of 100 profiles | 200; golden check passed |
| Semantic canary | 200; predictions match |

**SAVED REAL HTTP RESPONSE**


```
{
  "segment": "Segment 1",
  "cluster_id": 0,
  "distance": 4.1435,
  "uncertain": false,
  "model_version": "abfd88e893ca"
}
```


Exactly 24 named, finite ratings in [0,5]. A valid uncertain profile returns a null segment.

Source: output/presentation/evidence/api-demo.json; service/schemas.py

# FAILED TESTS BLOCK DEPLOYMENT

## NORMAL RUN

Tests: success

Deploy: success

Real HTTP smoke: passed

Open GitHub Actions run 38173375457

## DELIBERATE FAILURE

Tests: failure

Deploy: skipped

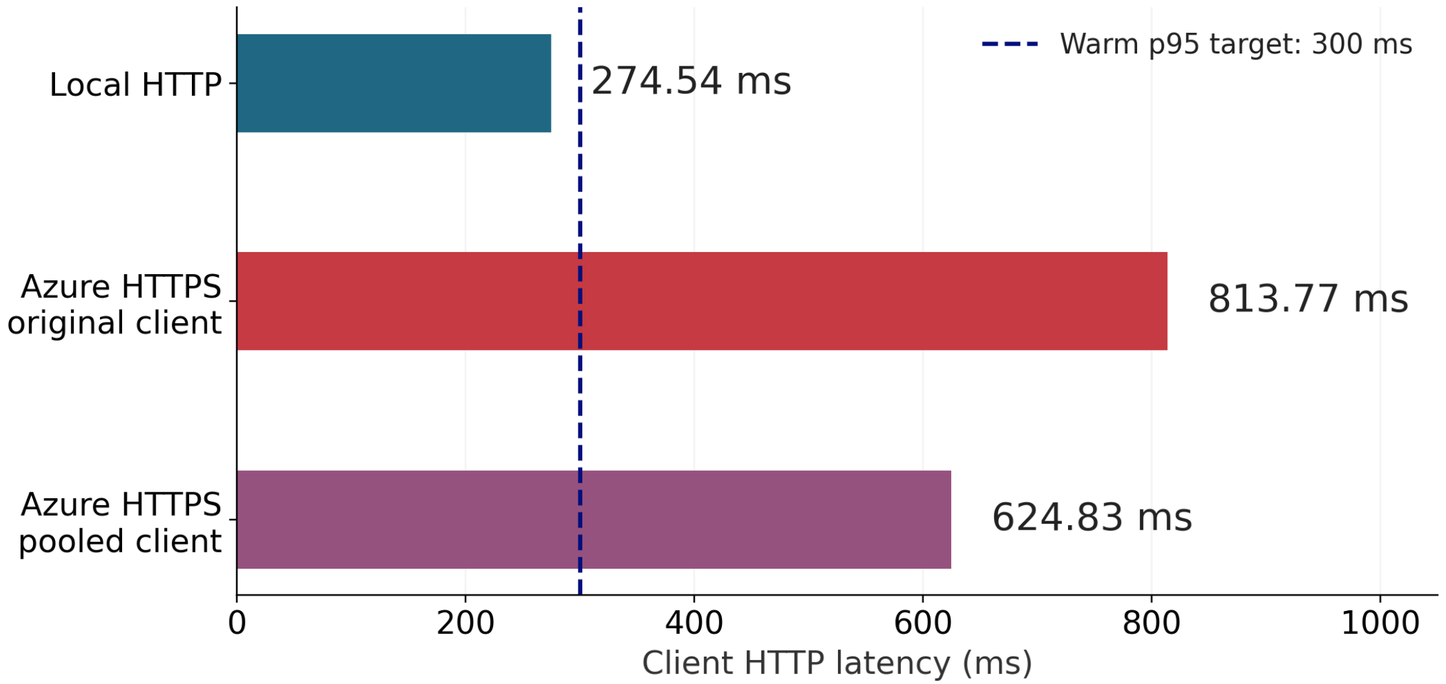
Failed code was not shipped

Open GitHub Actions run 38173380800

Same tested source commit: 9fea51a | OIDC login + limited roles | Immutable image

Source: reports/ci-runs.json; reports/ci-deploy-result.json; .github/workflows/ci.yml

# THE API WORKS; CLOUD LATENCY MISSES OUR TARGET




## WARM TEST

200 requests

Concurrency: 10

Errors: 0 in all runs

## COLD REQUEST

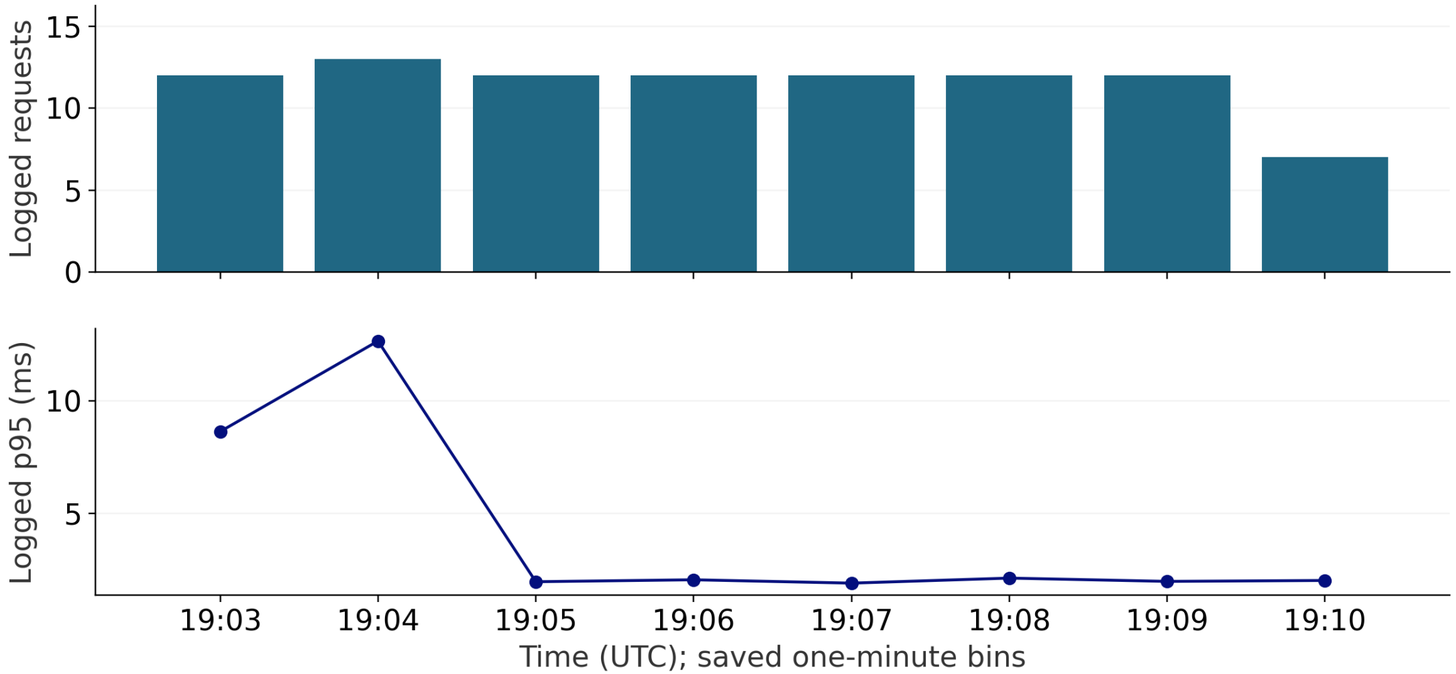
3.82 seconds

After verified zero replicas

One WSL HTTPS sample

Source: reports/serving-recheck/local-load.json; azure-load.json; pooled-load.json; finish-cold.json

# MONITOR REQUESTS AND CHECK PREDICTION MEANING




## DASHBOARD

MLOP Travel Style

Requests, errors, latency

Model version in logs

## ACTUAL ALERT

Semantic failure detected

Alert fired, then resolved

Email is not configured

Saved Log Analytics results; application timing, not client HTTPS latency.

Source: reports/azure-telemetry.json; finish-monitor.json; finish-alert.json

# A WRONG SEGMENT MAPPING TRIGGERED DETECTION AND RECOVERY

Controlled fault in the separate staging app: wrong cluster-to-segment mapping.

**INJECT**

16:54:22 UTC

Wrong mapping

**DETECT**

16:54:52 UTC

Canary: 503

**ALERT**

16:56:39 UTC

Actual alert fired

**RECOVER**

17:02:08 UTC

Canary: 200

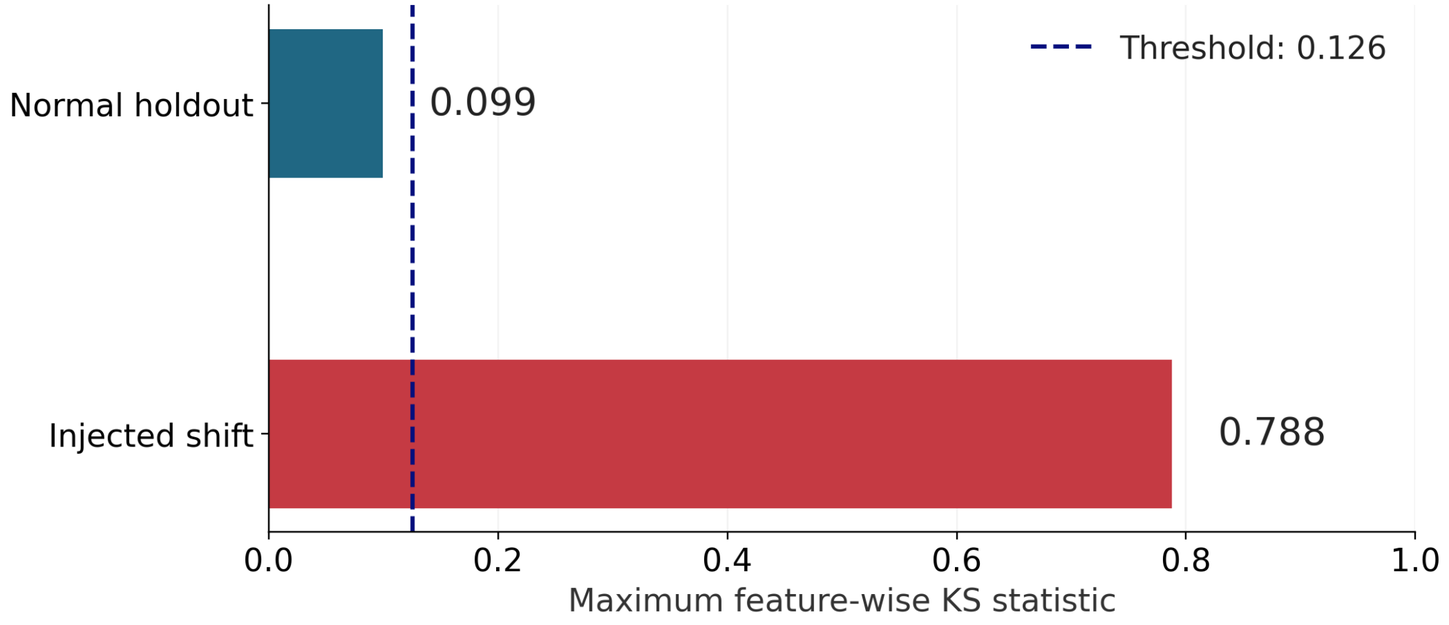
Detection to confirmed recovery: 7 min 16.4 sec, including the failed first recovery attempt.

Fix: restore config, restart staging, confirm the canary twice, and keep a regression test.

Saved failure evidence. Arrows show event order, not equal time gaps.

Source: reports/finish-failure.json; finish-alert.json; tests/test_api.py

# A CONTROLLED SHIFT IS DETECTED OFFLINE




## CONTROLLED TEST

Reference: 1,908 rows

Recent sample: 200 rows

100 calibration rounds

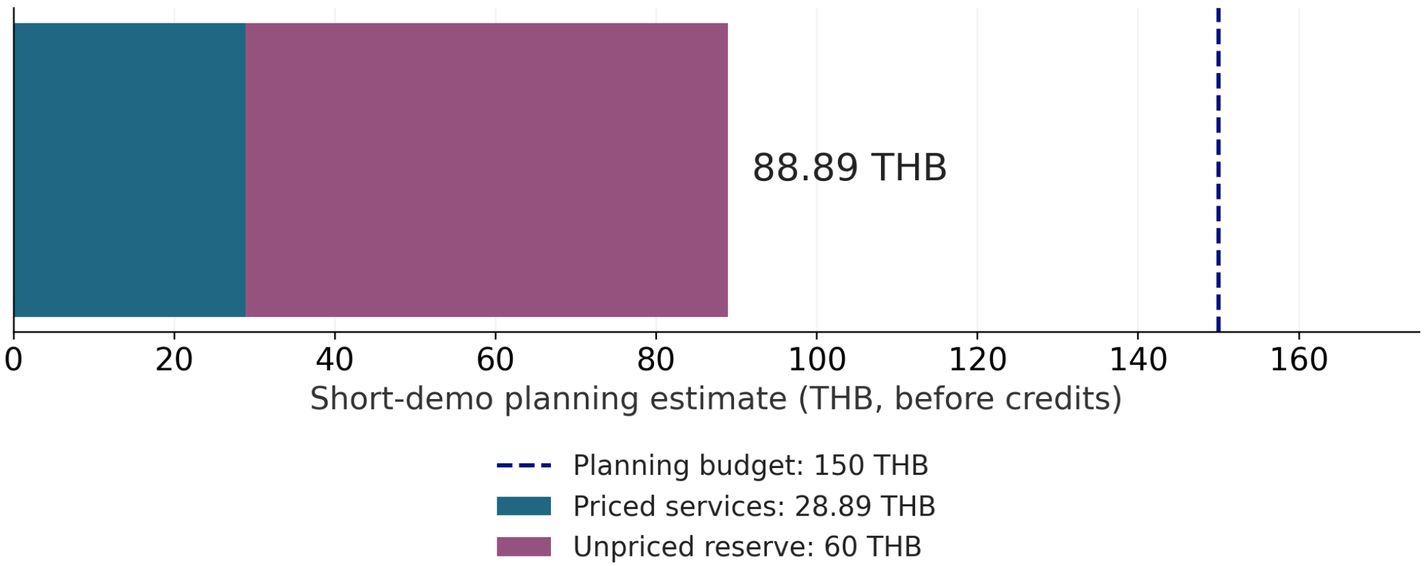
Shift: +1.5 in four ratings, clipped to [0,5].

Offline input drift only.

No concept-drift claim.

Source: reports/input-drift.json; monitoring/drift.py

# COST PER 1,000 PREDICTIONS: THE ESTIMATE AND ITS LIMITS




## PLANNED COST

8.89 THB / 1,000

Assume 10,000 predictions

One prediction per request

## REPORTED COST

0.00143104 USD / 1,000

841 observed profiles

Provisional; not a final bill

Assumptions: 40 THB/USD; training 1 h; app active 2 h; ACR 2 days. No credits applied.

Source: reports/azure-cost-plan.json; cost-per-1000.json; docs/COST_REPORT.md

# WHAT ANOTHER WEEK WOULD IMPROVE

| Priority | Next action |
| --- | --- |
| Cloud warm latency | Measure network vs. application time; retest p95 |
| Cost reconciliation | Recheck full cost and usage when billing updates |
| Live feature drift | Add rolling windows when current data is available |
| User quality | Review segment labels and test useful recommendations |

Current limits: provisional labels, unknown zero meaning, and no user-quality study.

Save evidence before cleanup. Remove cloud resources after the final demo.

Source: reports/finish-delivery.json; docs/DELIVERY_STATUS.md; docs/FINAL_STEPS.md

# REFERENCES

[1] S. Renjith, "Travel Review Ratings," UCI Machine Learning Repository, 2018.
DOI: 10.24432/C5C31Q. License: CC BY 4.0.

[2] ITCS355, "Capstone Project Brief," course/project/project-brief.md.
Presentation: 8 minutes + 5 minutes of questions.

[3] MLOP-PJ team, "Project reports and CI runs."
Data, registry, API, load, alert, recovery, drift, and cost evidence.

Dataset: https://doi.org/10.24432/C5C31Q

Repository: https://github.com/DewPerawis/MLOP-PJ (access required)

Source: UCI dataset page; course brief; project reports and CI runs

# THANK YOU